The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

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1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF).

2 The 30-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 30}$:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{emp (16)}} \mid \boldsymbol{g}_i^{\text{spatial (2)}} \mid \boldsymbol{p}_i^{\text{ppl (3)}} \mid \boldsymbol{r}_i^{\text{rsa (3)}} \mid \boldsymbol{s}_i^{\text{searle (3)}} \mid \boldsymbol{t}_i^{\text{deontic (3)}} \right] \in \mathbb{R}^{30} \tag{1}$$

3 Phase-Shift Stabilization Under Rigid Periodic Software Forcing

We formalize the mathematical safety perimeters enabling grid stabilization without modifying internal data center deep learning compilation batch variables.

3.1 The Fixed-Batch Periodic Driving Constraint

Let $\mathcal{W}_i(t)$ represent the rigid internal training workload execution function of data center $a_i \in \mathcal{A}$, operating with an unalterable continuous training stride of approximately 50 seconds. The resulting load profile generates an immutable, extensional periodic driving square wave:

$$P_{\text{AI},i}(t) = \text{Square_Wave}(\Omega_{\text{load}} \cdot t + \phi_i) \tag{2}$$

Uncoordinated initialization mandates that phase variables align constructively ($\phi_1 = \phi_2$), leading to resonant amplification that drives the second-order Kuramoto swing equations into period-doubling bifurcation cascades.

3.2 The External Phase-Shifting Gating Operator

To prevent grid collapse without intruding into the proprietary data center software stack, the MARL controller acts exclusively as an external access gatekeeper at the network boundary interface. Let $\mathcal{H}_{\text{CD}}$ represent the collision detection function tracking aggregate load derivatives ($\frac{dL}{dt}$). Upon detection of an imminent bus overlap event, the controller instantiates a randomized temporal phase-shifting gating operator $\mathcal{G}_{\Delta t}$:

$$P_{\text{AI},i}^{\text{gated}}(t) = \text{Square_Wave}\Big(\Omega_{\text{load}} \cdot (t - \Delta t_{\text{random}}) + \phi_i\Big) \tag{3}$$

The application of $\mathcal{G}_{\Delta t}$ introduces an artificial, out-of-band asymmetric delay that forces destructive interference across the active load profiles:

$$\phi_1 \neq \phi_2^{\text{gated}} \implies \int_t^{t+T} \Big(P_{\text{AI},1}(t) + P_{\text{AI},2}^{\text{gated}}(t)\Big)\, e^{-i\Omega_{\text{load}}t} dt \to 0 \tag{4}$$

By explicitly nullifying the low-frequency Fourier spectral magnitude within the critical resonant inter-area band ($0.1 - 20$ Hz), the destructive driving term is removed from the Kuramoto system manifold. The grid's natural damping d_i and mechanical turbine inertia m_i recover immediate dominance, dampening frequency deviations $\dot{\theta}_i(t)$ and pulling the coupled non-linear oscillators back into a stable, globally synchronized phase-locked steady state ($\circ$).